

VOCABULARY BUILDER – Space Exploration

Advantages and Disadvantages of Space Exploration – Key Words • LEVEL A

The words and phrases below appear in the article *Advantages and Disadvantages of Space Exploration*. Study the table carefully, then complete the gap-fill exercise in Part 2.

PART 1 – VOCABULARY TABLE

WORD / PHRASE	DEFINITION	EXAMPLE SENTENCE
captivate	to attract and hold someone's attention and interest completely	<i>Images from the James Webb Space Telescope continue to captivate scientists and the public alike.</i>
outer space	the region beyond Earth's atmosphere where the stars and planets are found	<i>Few humans have ever had the chance to travel to outer space.</i>
explosion	a sudden and violent burst of energy, often causing destruction	<i>The explosion of the Challenger shuttle shocked the world and halted the space programme.</i>
crew	a group of people who work together on a vehicle such as a ship, aircraft, or spacecraft	<i>The entire crew of seven lost their lives in the Challenger disaster.</i>
disaster	a sudden event that causes great harm, damage, or loss of life	<i>The Columbia disaster in 2003 was the second fatal accident in the shuttle programme.</i>
tragic loss	the death of people in a very sad and upsetting event	<i>Both shuttle accidents resulted in the tragic loss of highly skilled and dedicated crew members.</i>
mitigate	to reduce the severity or danger of something	<i>Lessons from past disasters have helped engineers mitigate the risks of future space missions.</i>
shuttle	a spacecraft designed to travel between Earth and space multiple times	<i>The space shuttle programme ran for 30 years before it was retired in 2011.</i>
lunar	relating to the moon	<i>The lunar module carried Neil Armstrong and Buzz Aldrin down to the surface of the moon.</i>
innovations	new ideas, methods, or inventions that bring about useful changes	<i>Space research has produced countless innovations that benefit everyday life on Earth.</i>
cost-effective	giving good value in relation to the money or time spent	<i>Some critics argue that space missions are not cost-effective compared to other priorities.</i>
satellite	an object sent into orbit around the Earth to collect and send back information	<i>Weather satellites help scientists monitor storms and predict natural disasters more accurately.</i>
atmosphere	the layer of gases surrounding the Earth or another planet	<i>NASA's satellites study the Earth's atmosphere to better understand climate change.</i>

WORD / PHRASE	DEFINITION	EXAMPLE SENTENCE
debris	pieces of waste or broken material left floating in space or scattered on a surface	<i>Thousands of pieces of debris are currently floating in low Earth orbit around the Earth.</i>
orbit	the curved path an object follows around a planet or star in space	<i>The International Space Station travels around Earth in a low orbit at 28,000 km per hour.</i>
worthwhile	worth the time, money, or effort spent on it	<i>Many people believe space exploration is worthwhile because of the knowledge and technology it produces.</i>

PART 2 – GAP-FILL EXERCISE

Complete the sentences using the correct word or phrase from the word bank below. You may need to change the form of the word. Some sentences need more than one answer.

WORD BANK

captivate	outer space	explosion	crew
disaster	tragic loss	mitigate	shuttle
lunar	innovations	cost-effective	satellite
atmosphere	debris	orbit	worthwhile

1	Space exploration continues to _____ millions of people, from scientists studying _____ to children dreaming of becoming astronauts.
2	The _____ of the Challenger _____ 73 seconds after launch, killing the entire _____ of seven.
3	The Columbia _____ in 2003 resulted in the _____ of another crew and led to a review of safety procedures.
4	Although dangerous, these accidents provided lessons that helped engineers _____ risks and develop safer _____ missions.
5	Critics argue that space programmes are not _____ — NASA's budget is over \$25 billion a year, which is a lot for taxpayers to fund.
6	However, space research has led to remarkable _____, including GPS, medical imaging, and _____ materials now used in buildings and stadiums.
7	NASA's _____ missions study the Earth's _____, helping scientists understand climate change and predict natural disasters.
8	Old _____ and equipment create serious _____ in _____ around the Earth — NASA has described this zone as an 'orbital graveyard'.

Most people believe the investment in space exploration is _____ because of the scientific knowledge and economic benefits it produces.

Write your answers here:

1. _____	2. _____	3. _____
4. _____	5. _____	6. _____
7. _____	8. _____	9. _____